Haemostasis

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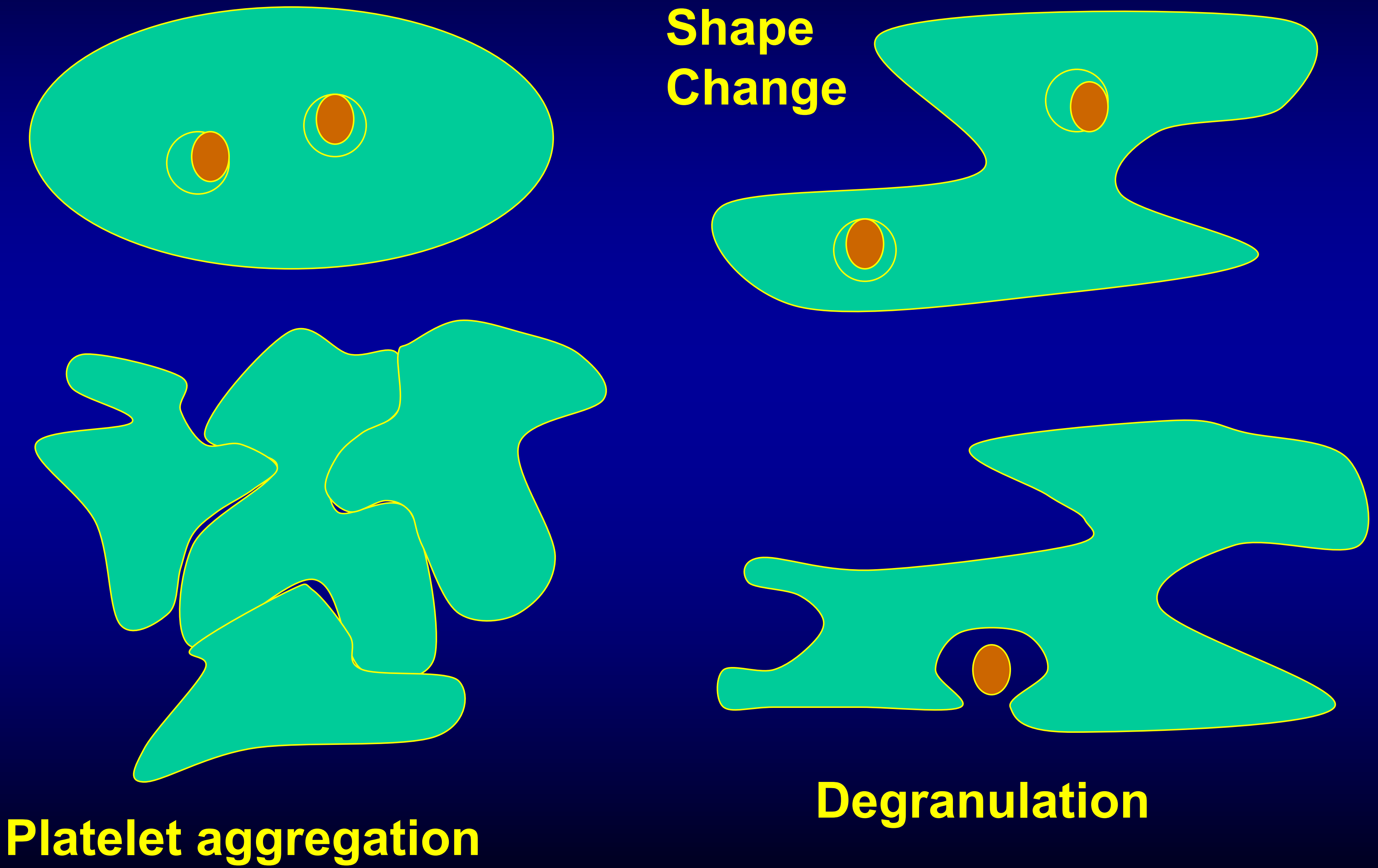
Three phases of haemostasis

1. Vasoconstriction
2. Platelet activation to the formation of a platelet plug
3. Activation of coagulation with the formation of a fibrin clot

Platelets

Phases of platelet activation



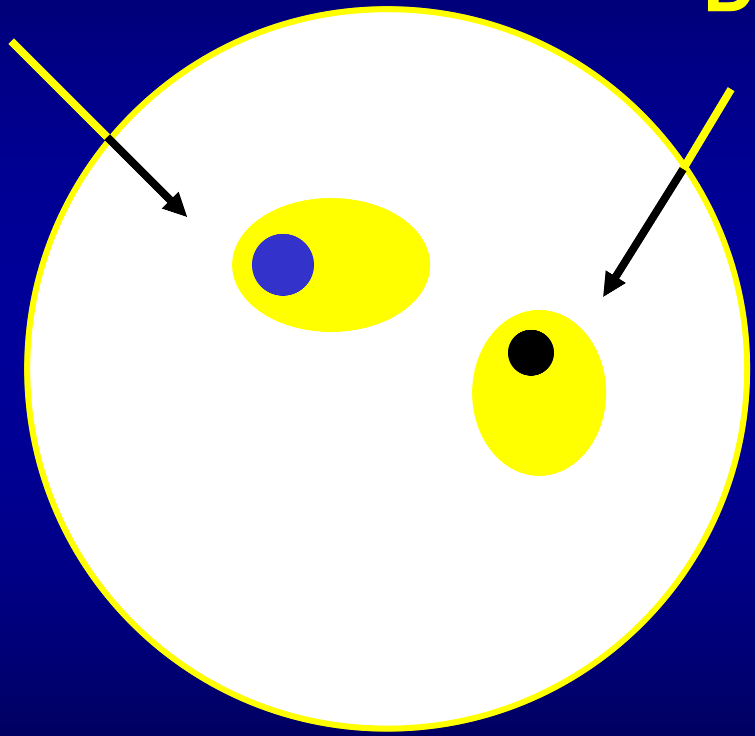
Main Agonists of platelet activation

Adenosine diphosphate (ADP)
Thrombin
Collagen
Adrenaline
Serotonin
Calcium ionophores

Platelet release (degranulation)

Alpha granule

fibrinogen
vWF
factor V
factor XI
HMWK



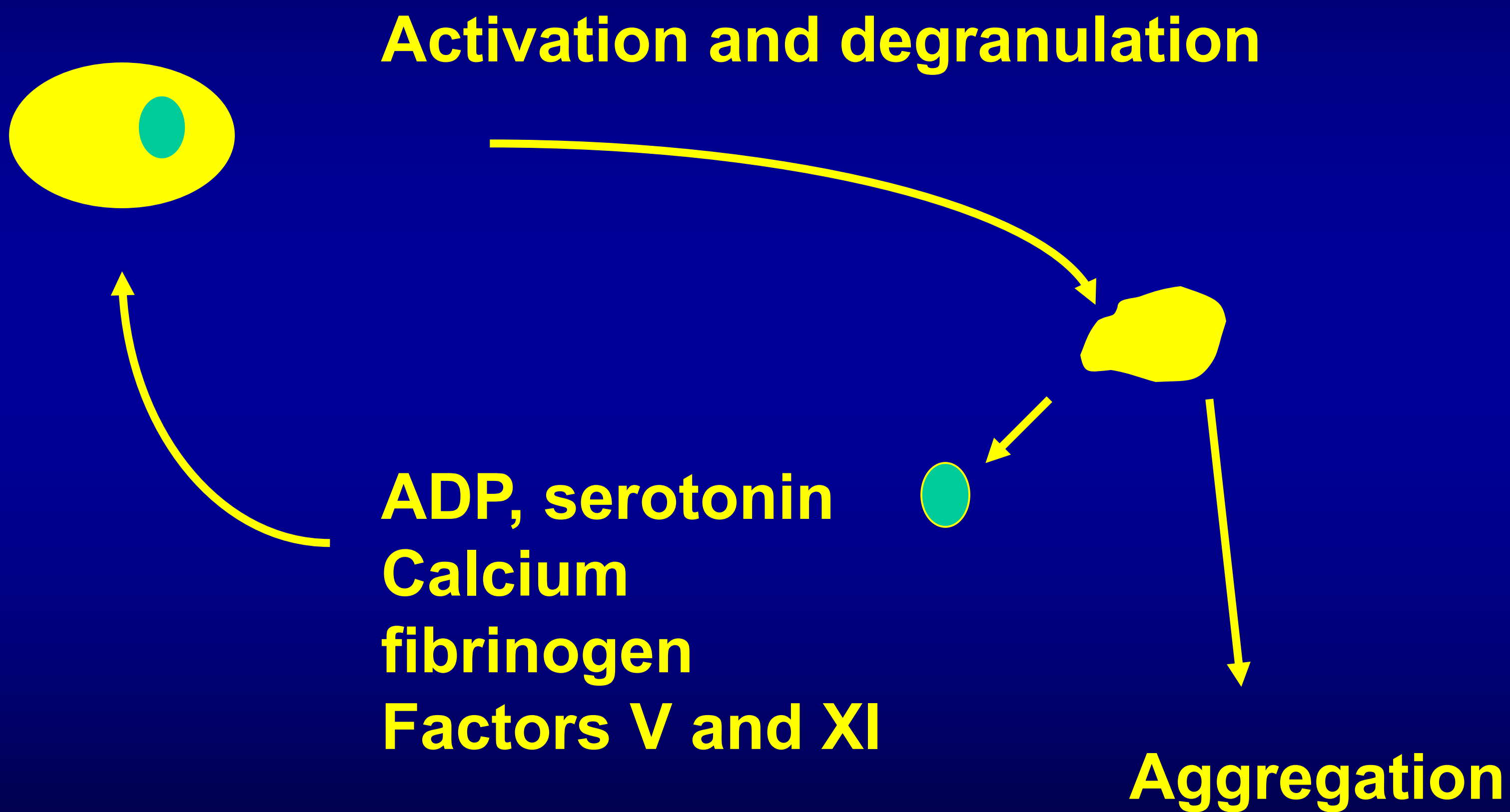
Dense granule

ADP
ATP
Serotonin
Calcium

Main Agonists of platelet activation

Adenosine diphosphate (ADP)
Thrombin
Collagen
Adrenaline
Serotonin
Calcium ionophores

Positive feedback of platelet activation



- Platelet adhesion

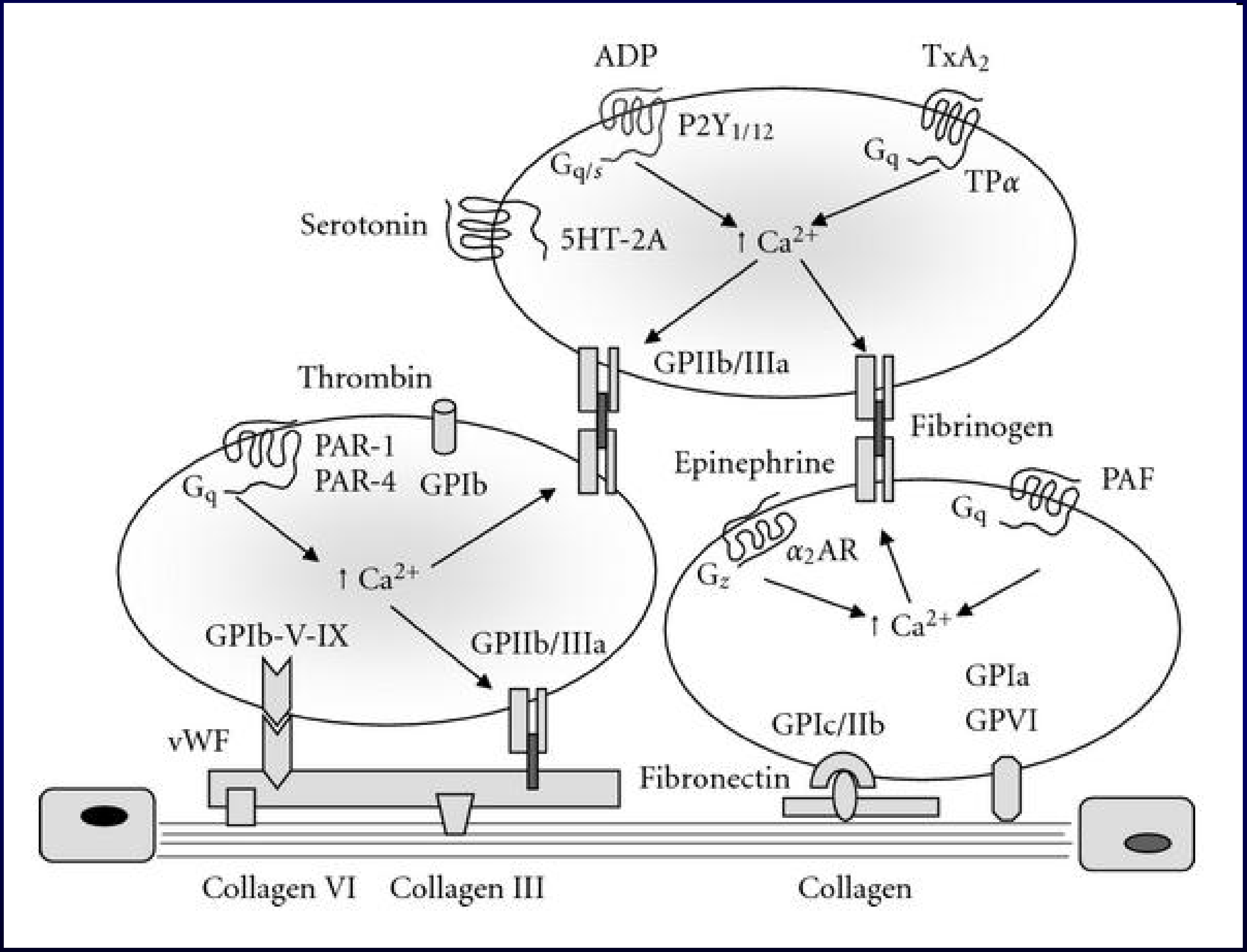
GPIa/IIa and GPVI – collagen

GPIb/IX/V – von Willebrand factor (vWF) – collagen

- Platelet aggregation

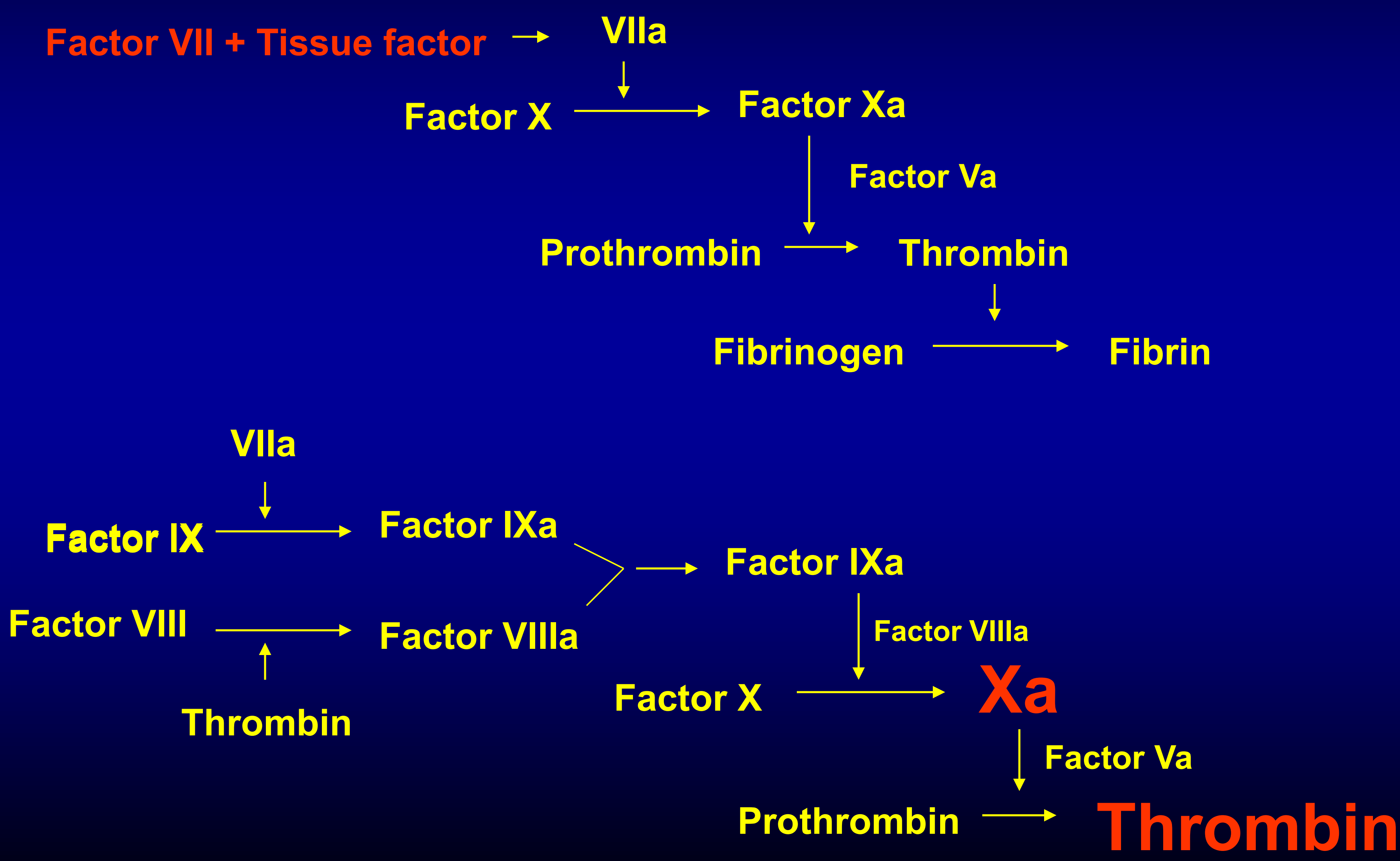
GPIIb/IIIa – fibrinogen – GPIIb/IIIa

Platelet adhesion and aggregation



Clotting factors

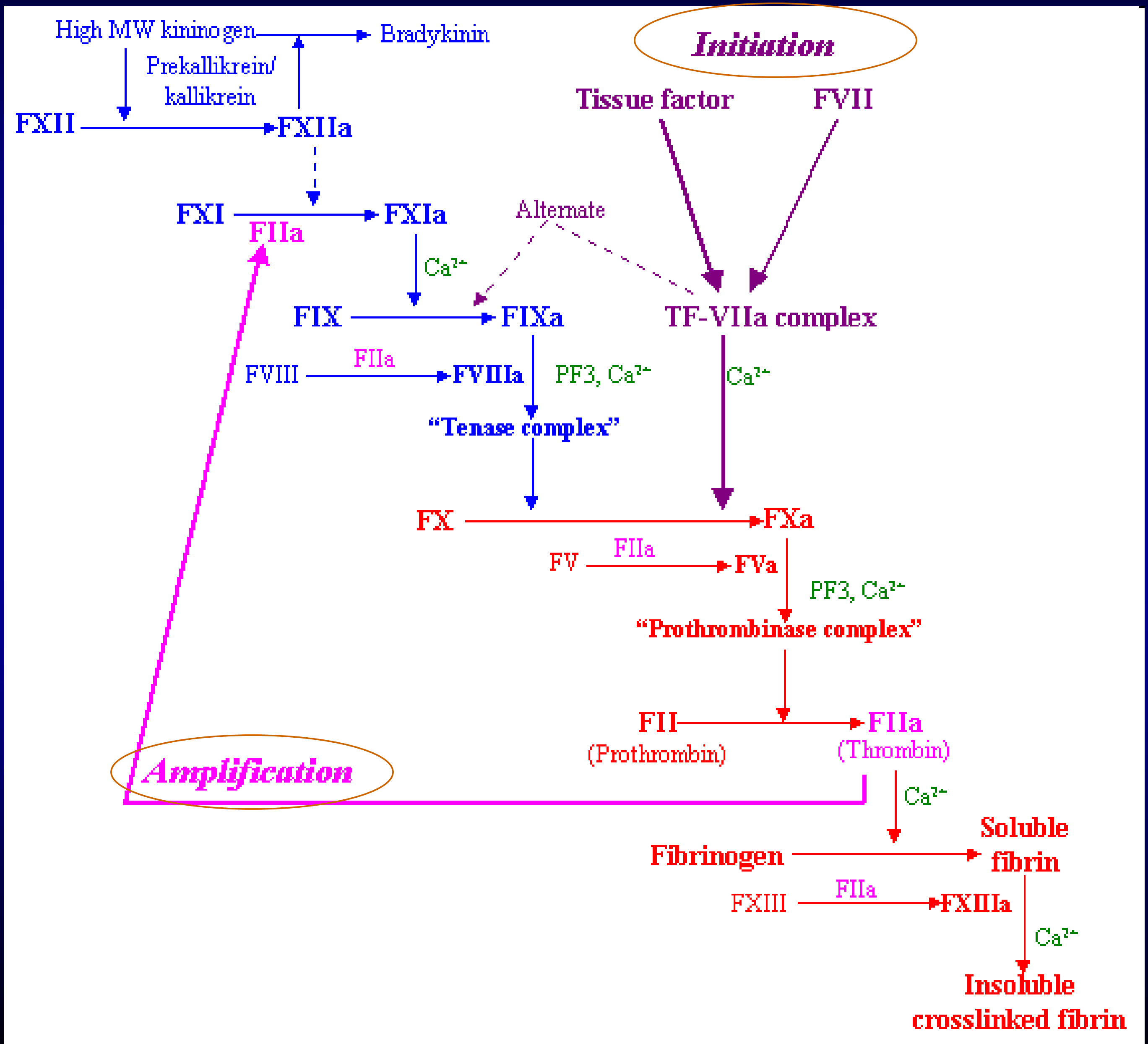
Activation of coagulation



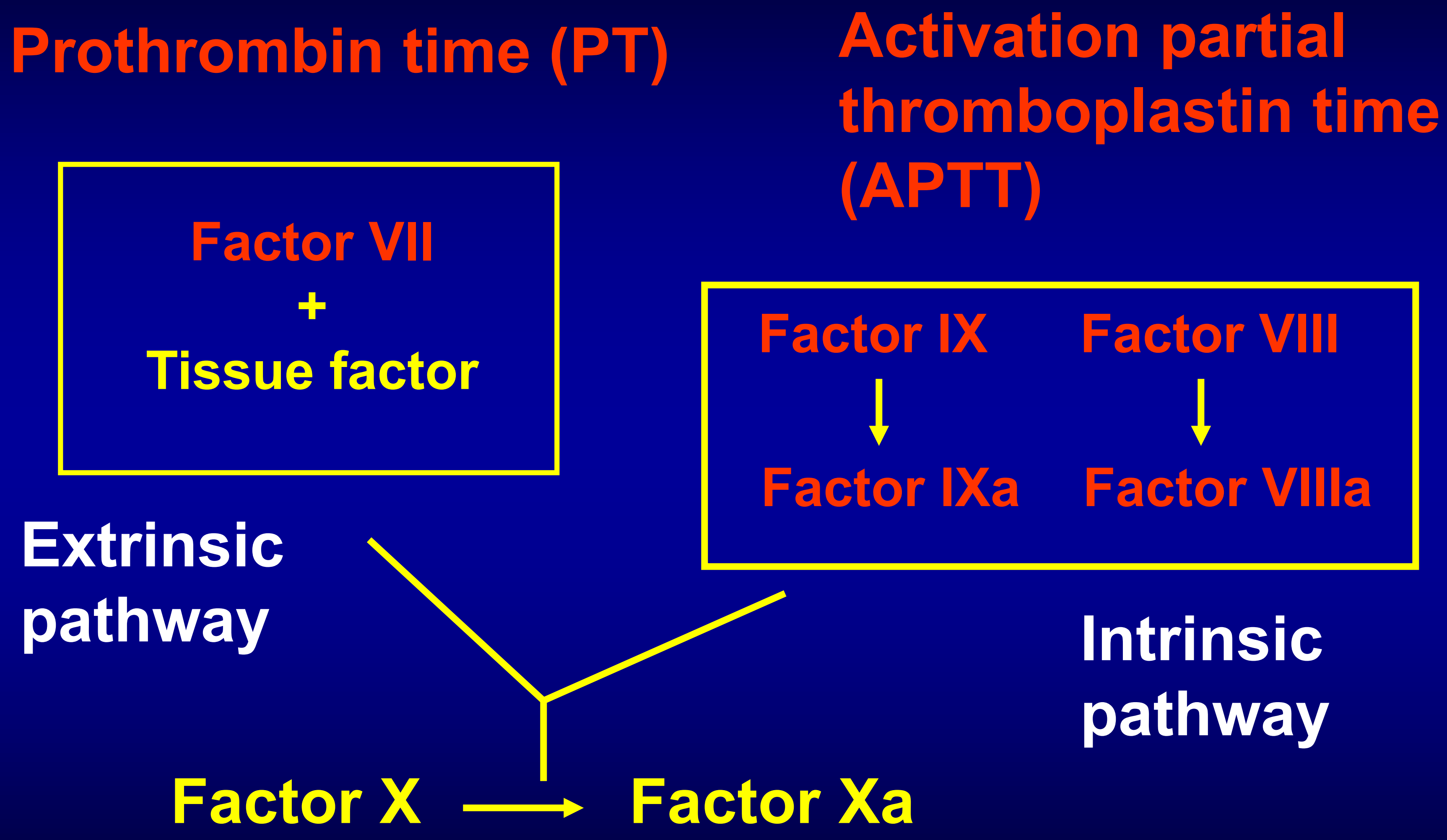
Principal steps in activation of coagulation

1. Factor VII is the key factor involved
2. Tissue factor is a principal factor involved in the activation of factor VII
3. Factor VIIa activates factor X
4. Factor X converts prothrombin to thrombin
5. Thrombin activates platelets, factor VIII and factor V and factor XI
6. Factor IXa/VIIIa generate large amount of THROMBIN via Xa/Va
7. Thrombin converts fibrinogen to fibrin
8. Thrombin is the central molecular involved in blood coagulation

Coagulation Cascade



Coagulation assays


Prothrombin time (PT)
Activation partial thromboplastin time (APTT)
Factor VII
+
Tissue factor
Factor IX
Factor VIII
Factor IXa
Factor VIIIa
Extrinsic pathway
Intrinsic pathway
Factor X
Factor Xa

Haemostatic disorders

Platelet defects

1. Quantitative : thrombocytopenia

    Decreased production : Bone marrow disease

    Increased destruction : immune (e.g. ITP, drugs)

    hypersplenism

2. Qualitative : Platelet function disorder

    Drugs : aspirin

    non steroidal anti-inflammatory agents

    Acquired : renal failure

    Inherited : very uncommon e.g. Bernard-Soulier

Coagulation defects

1. Acquired coagulation disorders

Liver disease : Vitamin K deficiency in jaundice
Decrease production in liver failure

Drugs : anti-coagulants

2. Inherited coagulation disorders

Haemophilias (A, B and other haemophilias)
von Willebrand disease (vWD)

Clinical and laboratory features of abnormal bleeding

Types of spontaneous bleeding	Platelet disorders	Coagulation disorders
1. Subcutaneous bleeding		
Petechiae (< 2 mm)	Yes	No
Purpura (2 -10 mm)	Yes	Usually no
Ecchymosis (> 1 cm)	Yes	Yes
2. Mucosal bleeding	Yes	Usually no
3. Intracranial bleeding	Yes	Yes
4. Retinal haemorrhage	Yes	Usually no
4. Intramuscular and Intra-articular bleeding	Usually no	Yes

Patterns of Bleeding : platelet vs coagulation

Laboratory tests of haemostasis

Test	Clinical application
1. Skin bleeding time	Nil
2. Platelet count	Very useful
3. Platelet function test	Not routine
4. Clotting time	Nil
5. Prothrombin time	Very useful
6. Activated partial thromboplastin time	Very useful
7. Other tests	Optional

Results of coagulation tests in various disorders

	PT	APTT
Jaundice (Vit K def)	↑	N
Warfarin	↑	N (↑)
Liver disease	↑	N (↑)
Heparin (UFH, and not LMWH)	N	↑
Haemophilia A / B	N	↑
Platelet disorders	N	N

Treatment options in haemostatic disorders

1. Platelet disorders (Qualitative / quantitative)

- Platelet transfusion
- Steroid / immunosuppressants (for ITP)
- Intravenous immunoglobulin / IVIg (for ITP)
- Thromobopoietin receptor agonists e.g. eltrombopag and romiplostim (for refractory ITP)

2. Coagulation disorders

- Vitamin K (jaundice, liver disease)
- Specific factors / factor concentrates (haemophilia, vWD)
- DDAVP (vWD)
- Fresh frozen plasma

Clinical situation one : Low platelet count

A. Repeat and confirm (citrate blood if needed)

B. Isolated thrombocytopenia or pancytopenia

    Pancytopenia : bone marrow failure

C. Isolated thrombocytopenia

    Increased destruction :

        chronic liver disease

        immune thrombocytopenia purpura

        drug induced

D. Assessment of bleeding risk

    CNS bleeding : fundoscopic examination

E. Treatment : platelet transfusion

    steroid / IVIg (for ITP)

Clinical situation two : Isolated prolonged PT

1. Repeat and confirm
2. Assess whether patient is jaundiced
   (Obstructive jaundice)
3. Assess whether patient has a liver disease
4. Drug : warfarin
5. Why obstructive jaundice ?
6. Treatment : parenteral vitamin K
7. If warfarin overdose :
   fresh frozen plasma
   vitamin K treatment

Clinical situation three : isolated prolonged APTT

1. Repeat and confirm
2. Commonest cause of prolonged APTT is heparin contamination (if blood taken from a catheter)
3. Lupus anti - coagulant (not associated with bleeding, associated with thrombosis *in-vivo*)
4. Man : haemophilia A / B (Sex linked recessive)
   von Willebrand disease (autosomal dominant)
   Woman : von Willebrand disease (autosomal dominant)
5. Treatment : haemophilia A / B, vWD
   Factor concentrates (plasma derived or recombinant)
   DDAVP for some vWD

Clinical situation four : Prolonged PT & APTT

1. Repeat and confirm
2. Measure fibrinogen concentration
3. Measure platelet concentration
4. “disseminated intravascular coagulopathy”
   ↑ PT, ↑ APTT, ↓ fibrinogen, ↓ platelets
   Red blood cell fragmentation
5. DIC with all these components is a very uncommon clinical situation
6. Treatment of DIC is to replenish the consumed coagulation factors
7. Reverse the underlying causative factor

THE END

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!

JC
WCS

Look for today's teacher
and the others you may have missed.
(HKU Portal login required)